BUSINESS INTELLIGENCE LAB – UNIT 2 | SESSION 7
EXPERIMENT 7

1. Aim

To integrate advanced analytics with Business Intelligence reports using Microsoft Power BI and generate business insights, strategic recommendations, and executive presentations that support organizational decision-making.

2. Objectives

1. To analyze business performance using existing Power BI reports and dashboards.
2. To evaluate sales performance across different geographical regions.
3. To analyze profitability by product category and customer segment.
4. To identify monthly and yearly sales trends.
5. To identify business strengths, weaknesses, and opportunities using analytical findings.
6. To support business conclusions with KPIs, charts, and data-driven evidence.
7. To prepare a Business Insight Report based on the analysis.
8. To develop strategic recommendations for improving business performance.
9. To communicate analytical findings through an executive presentation.

3. Tools / Software Required

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor
4. Microsoft PowerPoint
5. Microsoft Word or an equivalent document editor

Dataset Source: Global Superstore Dataset (Kaggle)
https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

4. Theory

Advanced Analytics: Advanced analytics involves examining business data to identify patterns, trends, relationships, and opportunities. It helps organizations move beyond basic reporting and make better-informed decisions.

Business Intelligence (BI): Business Intelligence refers to the processes and tools used to collect, analyze, and visualize data for monitoring performance and supporting organizational decision-making.

Trend Analysis: Trend analysis examines changes in business metrics over time. Monthly and yearly sales trends help management understand business growth, seasonal patterns, and changes in demand.

Comparative Analysis: Comparative analysis evaluates business performance across regions, product categories, and customer segments to identify strong and weak areas.

KPI Analysis: Key Performance Indicators (KPIs) are measurable values used to evaluate business performance. Examples include Total Sales, Total Profit, Total Orders, and Profit Margin.

Business Insight: A business insight is a meaningful conclusion drawn from data analysis that helps explain business performance and guides future actions.

Data-Driven Decision-Making: Data-driven decision-making uses verified analytical evidence rather than assumptions to select appropriate business strategies.

5. Business Scenario

Global Superstore management has observed that sales are increasing but profitability is not growing at the expected rate. The management team requires a detailed analytical review to identify the reasons behind this situation.

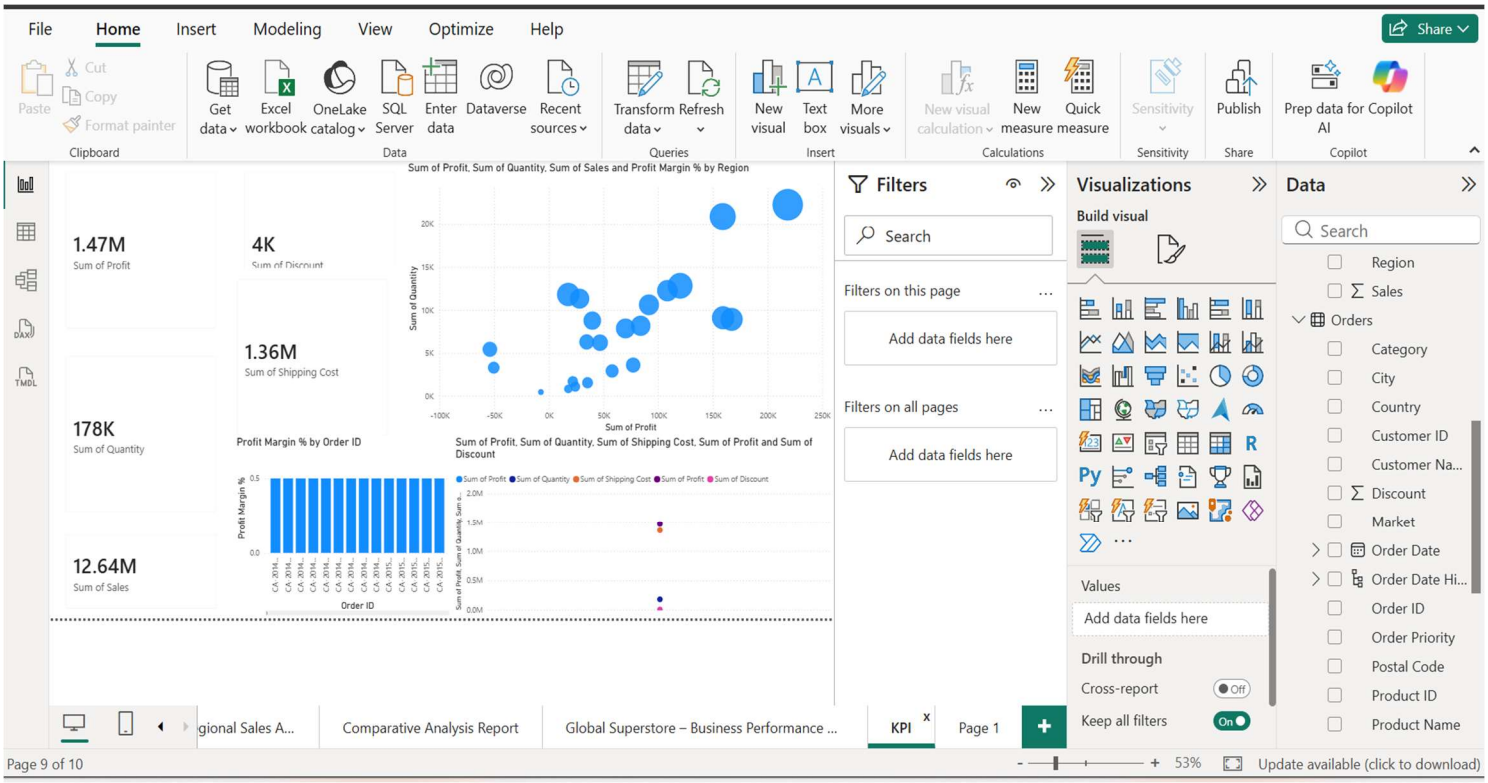
In this experiment, the role of a Business Intelligence Consultant is performed by analyzing existing Power BI dashboards, identifying business strengths and weaknesses, discovering improvement opportunities, and recommending suitable actions supported by data.

6. Procedure / Conduct

Part A: Analyze Existing Reports and Dashboards

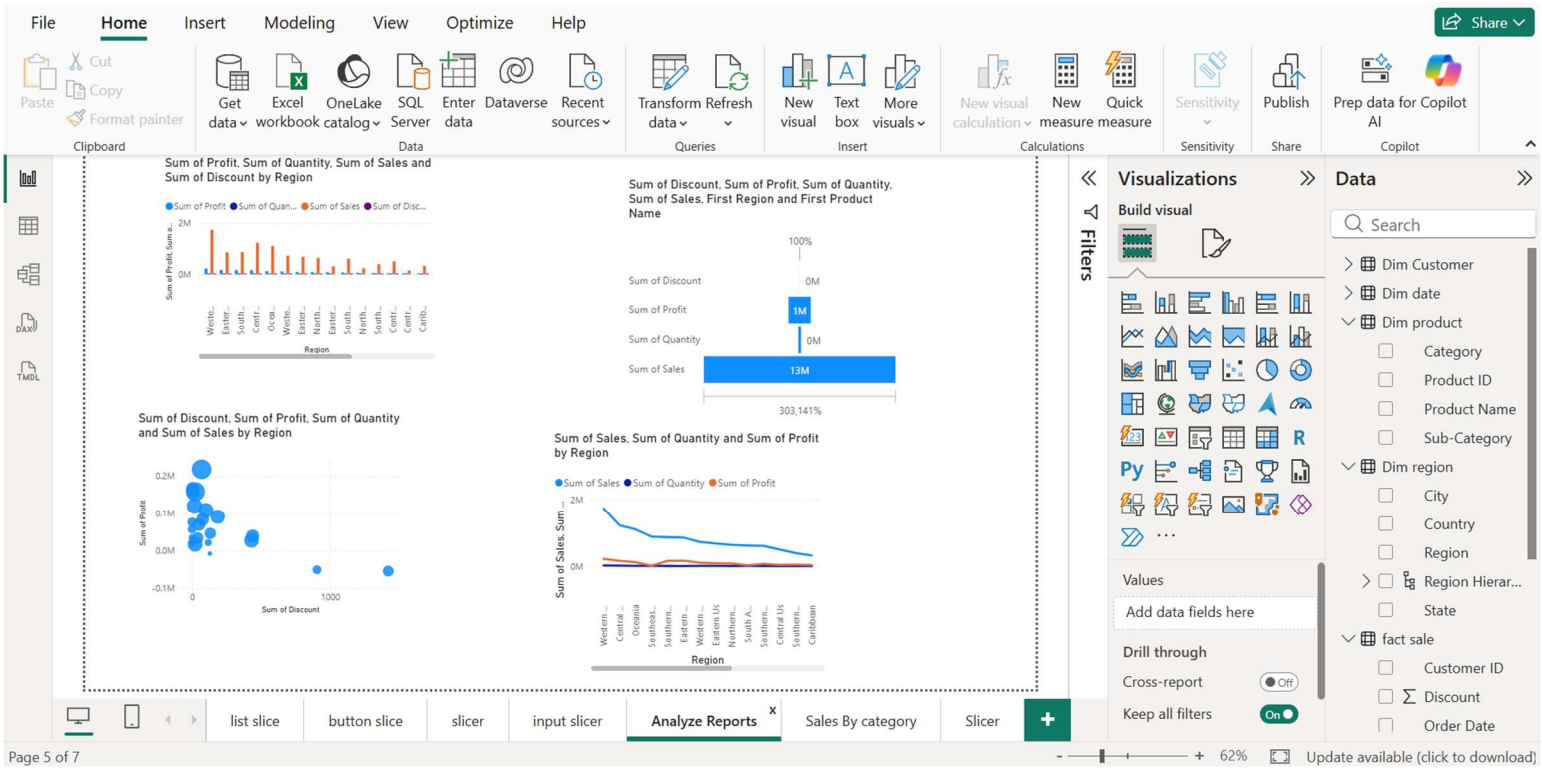
Step 1: Open Existing Power BI Reports

Opened the Power BI project developed in Experiments 5 and 6. The existing reports, KPI Cards, charts, slicers, and Business Scorecard were reviewed.



Step 2: Analyze Sales Performance by Region
Examined the Sales by Region chart to compare the sales performance of different geographical regions. The results were used to identify relatively strong and weak regional performance.

Step 3: Analyze Profitability by Product Category
Examined sales and profit performance across product categories. The analysis was used to identify categories contributing positively to profitability and categories requiring further investigation.



Step 4: Analyze Customer Segment Performance
Reviewed the Profit by Segment chart to compare the performance of Consumer, Corporate, and Home Office

segments.

Step 5: Analyze Monthly and Yearly Sales Trends

Examined the Monthly Sales Trend chart and the available date hierarchy to understand changes in sales over time.

Step 6: Document Observations

Recorded important findings from the charts, KPI Cards, and Business Scorecard. Findings were compared against the actual displayed values before drawing conclusions.

Part B: Business Problem Identification

Step 7: Identify Business Strengths

Business strengths were identified by examining areas with comparatively high sales, positive profitability, or strong customer-segment performance.

Examples of evidence to consider:

- Regions generating high sales and positive profit.
- Product categories contributing significantly to overall profit.

Step 8: Identify Business Weaknesses

Business weaknesses were identified by examining areas with low sales, weak profitability, or negative profit.

Examples of evidence to consider:

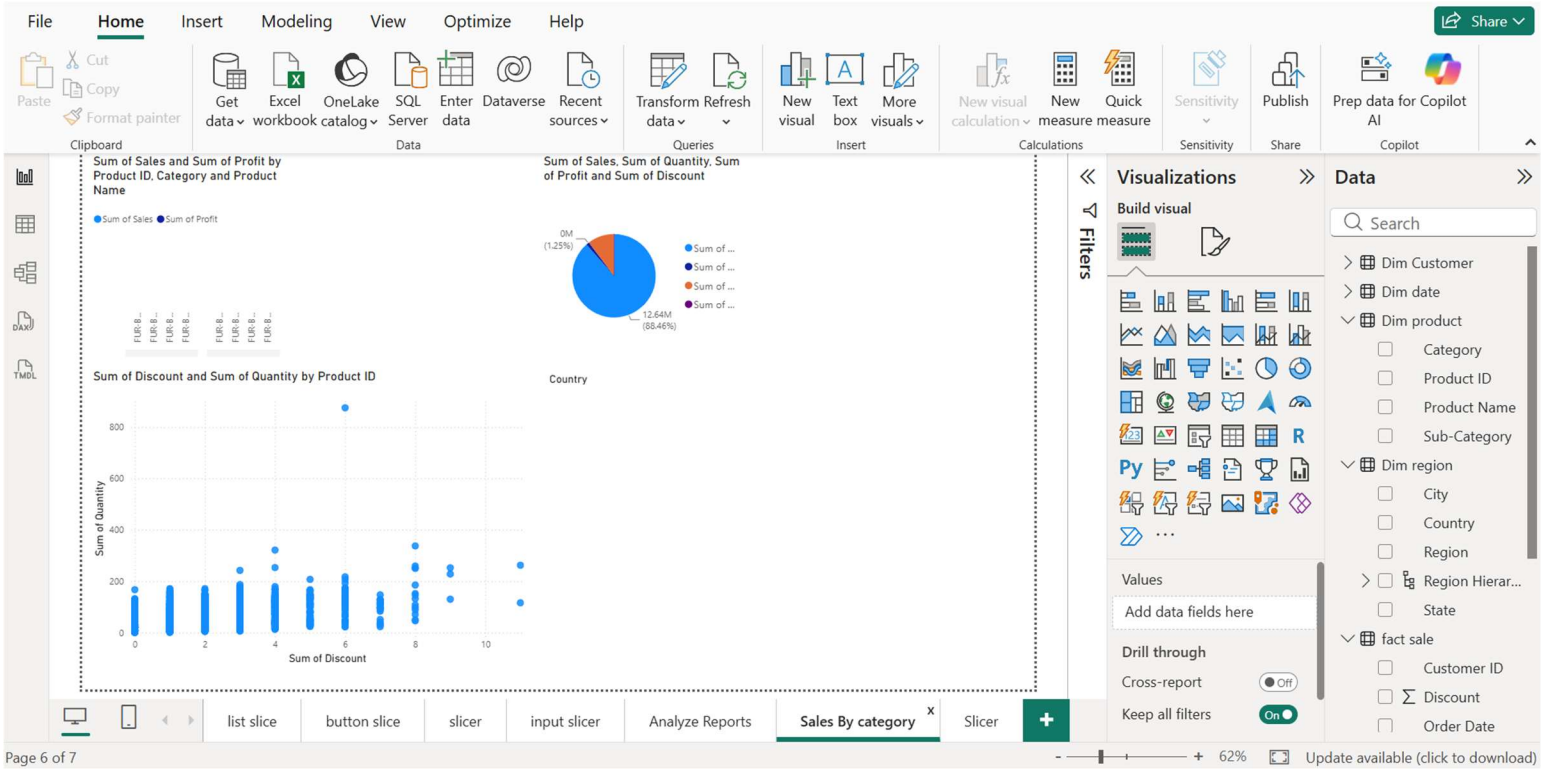
- Regions with comparatively low profit.
- Product categories or customer segments with negative or weak profitability.

Step 9: Identify Business Opportunities

Potential opportunities were identified by comparing the performance of regions, product categories, and customer segments.

Examples include:

- Improving profitability in underperforming regions.
- Reviewing pricing, discounts, and product costs in low-profit categories.
- Increasing sales of products or segments that demonstrate strong profitability.



Step 10: Support Findings with Analytical Evidence

The findings were supported using Power BI charts, KPI Cards, scorecard results, and numerical comparisons. Recommendations were based on verified data rather than assumptions.

Part C: Business Insight Report Preparation

Step 11: Prepare the Business Insight Report

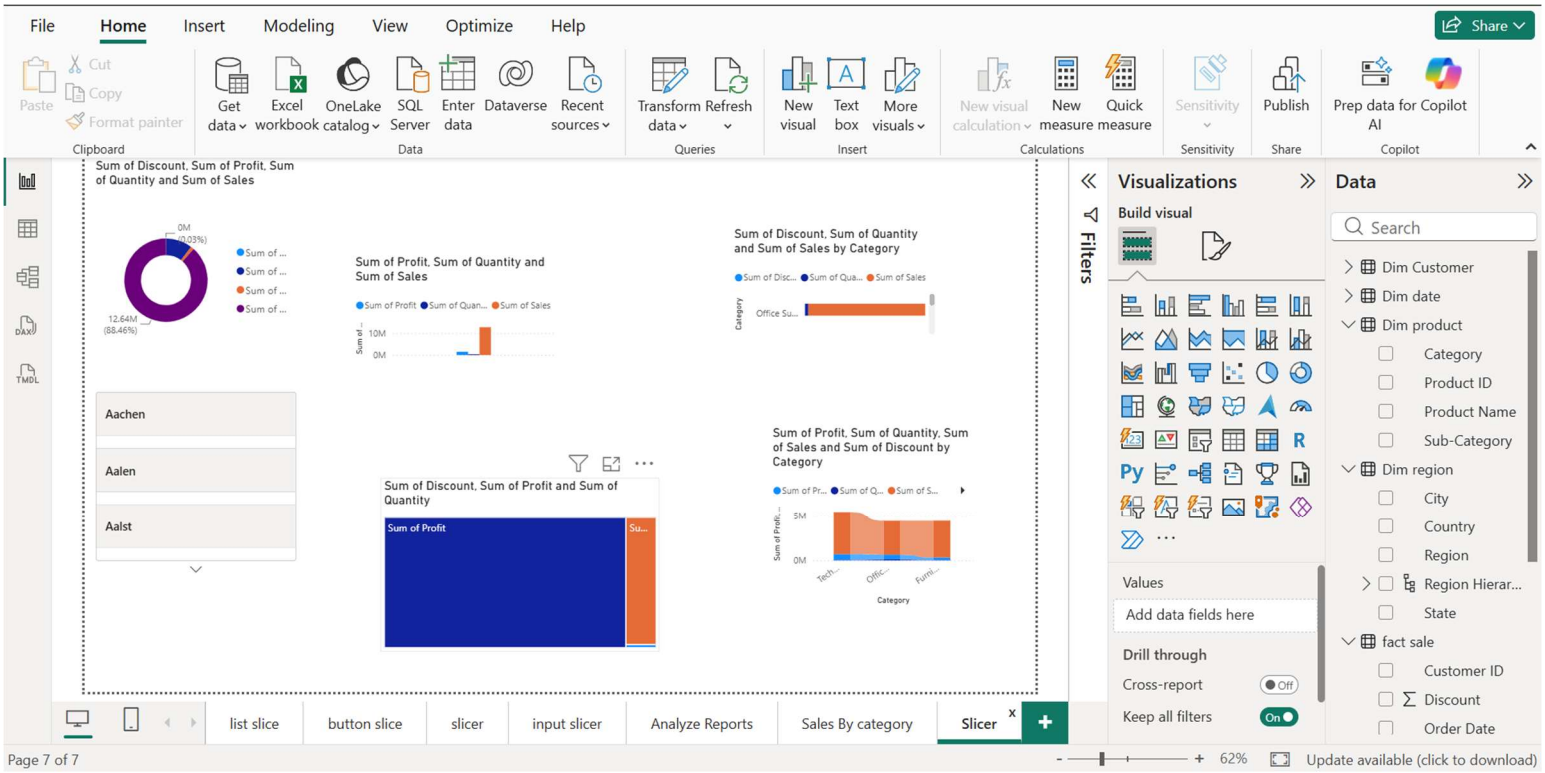
Prepared a structured report containing:

- Objective of Analysis
- Reports and Dashboards Used
- Key Findings
- Business Strengths
- Business Weaknesses
- Opportunities for Improvement
- Recommendations
- Conclusion

Step 12: Add Power BI Screenshots

Captured screenshots of the Executive Dashboard and relevant charts. The screenshots were included as supporting evidence with appropriate captions.

Part D: Executive Presentation



Step 13: Prepare the Executive Presentation

Prepared a presentation of 5–7 slides covering the introduction, dashboard overview, key findings, business issues, recommendations, and conclusion.

Step 14: Present Analytical Findings

The findings were organized in a clear sequence so that the audience could understand the business problem, the evidence, and the recommended actions.

Step 15: Justify Recommendations

Recommendations were linked to the analytical findings and supported using charts, KPIs, and numerical comparisons from Power BI.

8. Business Strengths

The following areas were considered when identifying business strengths:

1. **Strong Regional Performance:** Regions showing high sales and positive profit may represent important contributors to business performance.
2. **Profitable Product Categories:** Product categories generating higher profits may provide opportunities for sustainable business growth.

The final strengths should be supported by actual values from the Power BI reports.

9. Business Weaknesses

The following areas were considered when identifying business weaknesses:

1. **Low Regional Profitability:** Regions with relatively low or negative profit may require a review of costs, discounts, and sales strategies.
2. **Low Category Profitability:** Categories with weak profit margins or negative profit may require additional analysis of pricing and discount practices.

The final weaknesses should be confirmed using the actual dataset and dashboard findings.

10. Opportunities for Improvement

1. Improve the profitability of underperforming regions through targeted sales and cost-control strategies.
2. Review discounting and pricing practices for products with weak profitability.
3. Focus on products and customer segments that generate sustainable profits.
4. Monitor monthly and yearly sales trends to improve sales planning.
5. Use the Business Scorecard to track performance against clearly defined targets.

11. Recommendations

Based on the analytical approach, the following recommendations can be considered:

1. **Improve Profitability Monitoring:** Track Total Sales, Total Profit, and Profit Margin together instead of evaluating sales performance alone.
2. **Review Regional Performance:** Investigate regions with relatively low profitability and identify possible cost, pricing, or discount-related issues.
3. **Optimize Product Strategy:** Evaluate product categories according to both sales contribution and profit contribution.
4. **Monitor Customer Segments:** Compare the profitability of different customer segments to support more effective marketing and sales planning.
5. **Use Regular Trend Analysis:** Review monthly and yearly sales trends to identify changes in demand and support future planning.
6. **Track Business Targets:** Use the Power BI Business Scorecard to monitor actual performance against appropriate business targets.
7.

These recommendations should be prioritized according to the actual findings from the dataset.

12. Conclusion

In this experiment, advanced analytics techniques were integrated with existing Business Intelligence reports and dashboards developed using Microsoft Power BI. Regional sales, product-category profitability, customer-segment performance, and sales trends were examined to understand organizational performance.

The analysis provided a structured approach for identifying business strengths, weaknesses, and opportunities. The findings were used to develop recommendations and prepare an executive presentation. The experiment demonstrated how analytical reports and dashboards can support evidence-based business decisions.

13. Learning Outcomes

1. Learned to apply analytical techniques to business reports and dashboards. (CO3-BT3)
2. Learned to analyze organizational performance using KPIs and visualizations. (CO3-BT4)
3. Developed the ability to identify business strengths, weaknesses, and opportunities.
4. Learned to generate business insights from analytical findings. (CO5-BT6)
5. Learned to formulate strategic recommendations based on data-driven evidence. (CO5-BT6)
6. Improved the ability to communicate analytical findings through professional reports and presentations. (CO5-BT6)